



Techniques de sécurité informatique
420-4D5-MO

Travail pratique 2 – Partie 3

Présenté à : Mohammed El Koutbi
Remis par: Roméo Kouémeni Tongambou

19 mai 2026

PfSense (3 points)	4
Routeur Cisco avec ACL (5 points)	17
Snort avec PfSense (2 points)	28

Travail pratique 2 – Partie 3 (10%)

Filtrage ACL et PfSense

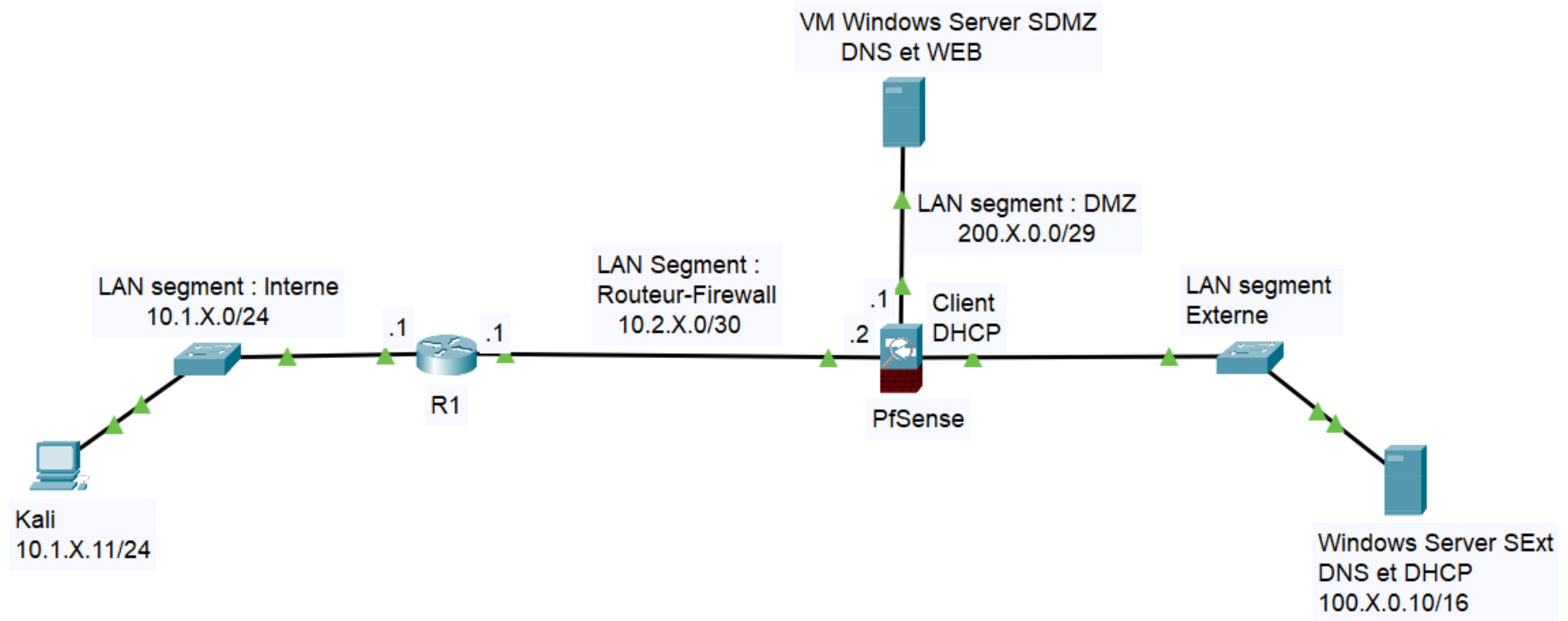
Le travail pratique 2 comporte 3 parties :

- Partie 1 : Attaque DOS et Authentification sécurisée (10 points)
- Partie 2 : Filtrage ACL et WiFi sécurisé (10 points)
- Partie 3 : Filtrage ACL et PfSense (10 points).

Partie 3 : Filtrage ACL et PfSense (10 points).

La partie 3 du TP2 est à soumettre le 15 mai avant minuit.

Réalisez la maquette ci-après. Votre X est celui donné au labo1.



PfSense (3 points)

- 1- Installez et configurez une VM PfSense comme indiqué sur le schéma ci-dessus ?

```
exit
VMware Virtual Machine - Netgate Device ID: 1d0e55160a055aefff6b

*** Welcome to pfSense 2.7.2-RELEASE (amd64) on pfSense ***

WAN (wan)      -> em0      -> v4/DHCP4: 100.64.0.1/16
LAN (lan)      -> em1      -> v4: 10.2.64.2/30
OPT1 (opt1)    -> em2      -> v4: 200.64.0.1/29

0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults  13) Update from console
5) Reboot system              14) Enable Secure Shell (sshd)
6) Halt system                15) Restore recent configuration
7) Ping host                  16) Restart PHP-FPM
8) Shell

Enter an option: █
```

2- Configurez la politique de filtrage suivante sur le routeur PfSense :

- a. Autorisez toutes les communications depuis le LAN (Routeur-Firewall) vers la DMZ ?

Firewall / Rules / Edit

Edit Firewall Rule

Action Pass
 Choose what to do with packets that match the criteria specified below.
 Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled ☐ Disable this rule
 Set this option to disable this rule without removing it from the list.

Interface LAN
 Choose the interface from which packets must come to match this rule.

Address Family IPv4
 Select the Internet Protocol version this rule applies to.

Protocol Any
 Choose which IP protocol this rule should match.

Source

Source ☐ Invert match LAN subnets Source Address /

Destination

Destination ☐ Invert match DMZ subnets Destination Address /

Extra Options

Log ☐ Log packets that are handled by this rule
 Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the [Status: System Logs: Settings](#) page).

Description LAN Routeur-Firewall vers DMZ
 A description may be entered here for administrative reference. A maximum of 52 characters will be used in the ruleset and displayed in the firewall log.

Advanced Options Display Advanced

Save

b. Autorisez toutes les communications depuis le LAN (Routeur-Firewall) vers le réseau externe (LAN Externe n'est pas 'Bridged') ?

Edit Firewall Rule

Action

Pass

Choose what to do with packets that match the criteria specified below.

Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled

☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface

LAN

Choose the interface from which packets must come to match this rule.

Address Family

IPv4

Select the Internet Protocol version this rule applies to.

Protocol

Any

Choose which IP protocol this rule should match.

Source

Source

☐ Invert match

LAN subnets

Source Address

/

Destination

Destination

☐ Invert match

WAN subnets

Destination Address

/

Extra Options

Log

☐ Log packets that are handled by this rule

Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the [Status: System Logs: Settings](#) page).

Description

LAN Routeur-Firewall vers Externe

A description may be entered here for administrative reference. A maximum of 52 characters will be used in the ruleset and displayed in the firewall log.

Advanced Options



Display Advanced

- c. Autorisez toutes les communications SSH, DNS et WEB depuis le réseau externe vers la DMZ ?

Action

Pass

Choose what to do with packets that match the criteria specified below.

Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface

WAN

Choose the interface from which packets must come to match this rule.

Address Family

IPv4

Select the Internet Protocol version this rule applies to.

Protocol

TCP

Choose which IP protocol this rule should match.

Source**Source**☐ Invert match

WAN subnets

Source Address

/

▼



Display Advanced

The **Source Port Range** for a connection is typically random and almost never equal to the destination port. In most cases this setting must remain at its default value, **any**.

Destination**Destination**☐ Invert match

Address or Alias

200.64.0.2

/

▼

Destination Port Range

SSH (22)

From

Custom

SSH (22)

To

Custom

Specify the destination port or port range for this rule. The "To" field may be left empty if only filtering a single port.

Extra Options**Log**☐ Log packets that are handled by this rule

Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the [Status: System Logs: Settings](#) page).

Description

Externe vers SDMZ SSH

A description may be entered here for administrative reference. A maximum of 52 characters will be used in the ruleset and displayed in the firewall

Edit Firewall Rule

Action

Pass

Choose what to do with packets that match the criteria specified below.

Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled

☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface

WAN

Choose the interface from which packets must come to match this rule.

Address Family

IPv4

Select the Internet Protocol version this rule applies to.

Protocol

UDP

Choose which IP protocol this rule should match.

Source

Source

☐ Invert match

WAN subnets

Source Address

/

 Display Advanced

The **Source Port Range** for a connection is typically random and almost never equal to the destination port. In most cases this setting must remain at its default value, any.

Destination

Destination

☐ Invert match

Address or Alias

200.64.0.2

Destination Port Range

DNS (53)

DNS (53)

From

Custom

To

Custom

Specify the destination port or port range for this rule. The "To" field may be left empty if only filtering a single port.

Extra Options

Log

☐ Log packets that are handled by this rule

Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the [Status: System Logs: Settings](#) page).

Description

Externe vers SDMZ DNS UDP

A description may be entered here for administrative reference. A maximum of 52 characters will be used in the ruleset and displayed in the firewall log.

Edit Firewall Rule

Action

Pass

Choose what to do with packets that match the criteria specified below.

Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable) is sent back to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled

☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface

WAN

Choose the interface from which packets must come to match this rule.

Address Family

IPv4

Select the Internet Protocol version this rule applies to.

Protocol

TCP

Choose which IP protocol this rule should match.

Source

Source

☐ Invert match

WAN subnets

Source Address

Display Advanced

The Source Port Range for a connection is typically random and almost never equal to the destination port, its default value, any.

Destination

Destination

☐ Invert match

Address or Alias

200.64.0.2

Destination Port Range

DNS (53)

From

Custom

DNS (53)

To

Custom

Specify the destination port or port range for this rule. The "To" field may be left empty if only filtering a single port.

Extra Options

Log

☐ Log packets that are handled by this rule

Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, check the [Status: System Logs: Settings](#) page).

Description

Externe vers SDMZ DNS TCP

A description may be entered here for administrative reference. A maximum of 52 characters will be used in the log.

Advanced Options

Display Advanced

Save

Edit Firewall Rule

Action

Pass

Choose what to do with packets that match the criteria specified below.

Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled

☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface

WAN

Choose the interface from which packets must come to match this rule.

Address Family

IPv4

Select the Internet Protocol version this rule applies to.

Protocol

TCP

Choose which IP protocol this rule should match.

Source

Source

☐ Invert match

WAN subnets

Source Address

 Display Advanced

The **Source Port Range** for a connection is typically random and almost never equal to the destination port. In most cases this is its default value, any.

Destination

Destination

☐ Invert match

Address or Alias

200.64.0.2

Destination Port Range

HTTP (80)

From

Custom

HTTP (80)

To

Custom

Specify the destination port or port range for this rule. The "To" field may be left empty if only filtering a single port.

Extra Options

Log

☐ Log packets that are handled by this rule

Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote log. See the [Status: System Logs: Settings](#) page).

Description

Externe vers SDMZ WEB

A description may be entered here for administrative reference. A maximum of 52 characters will be used in the ruleset and displayed in the log.

Advanced Options

 Display Advanced

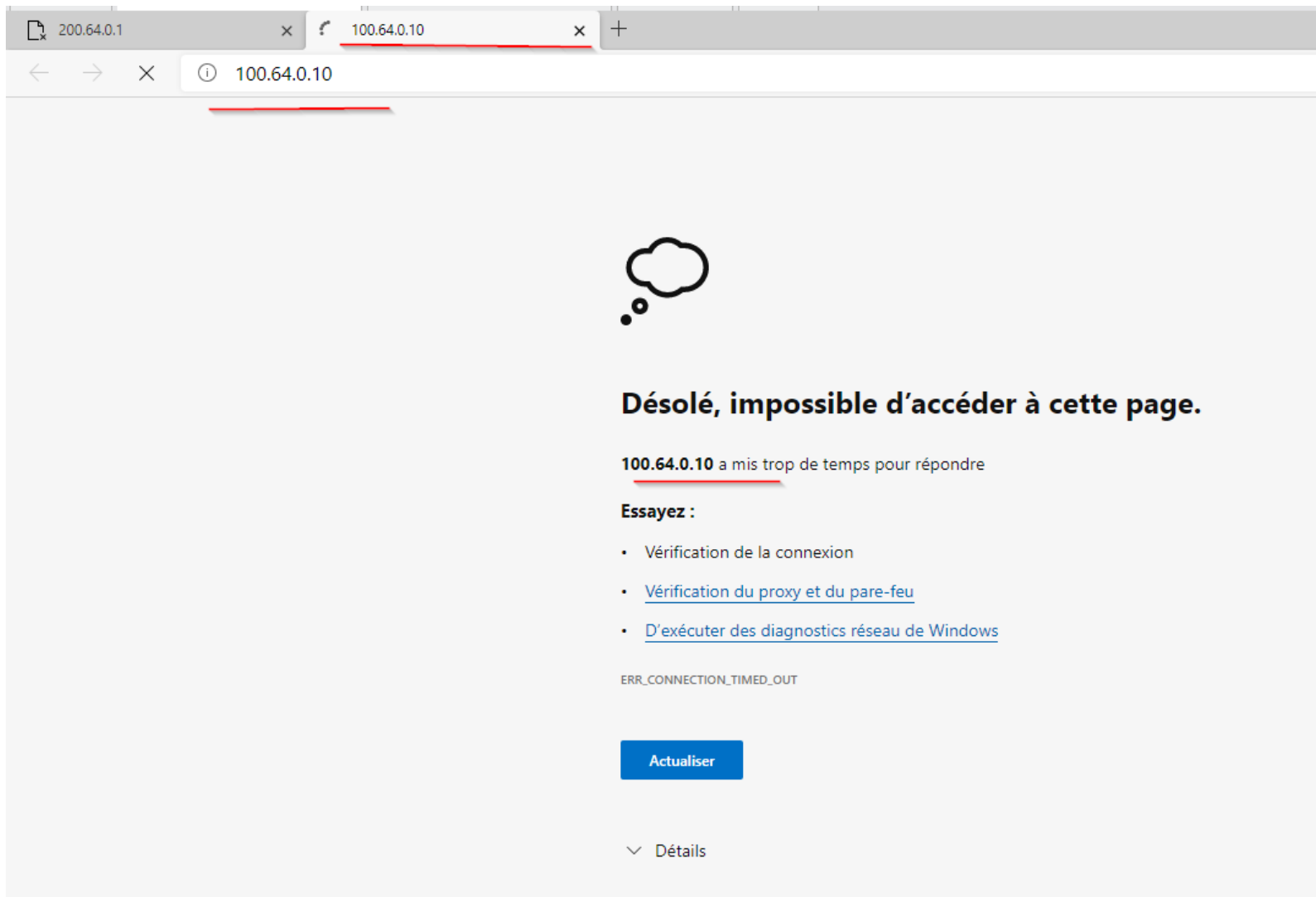
 Save

3- Testez les communications (ping, SSH et WEB) depuis la VM SDMZ vers la VM SExt ?

```
C:\Users\Administrateur>ping 100.64.0.10

Envoi d'une requête 'Ping' 100.64.0.10 avec 32 octets de données :
Délai d'attente de la demande dépassé.
Délai d'attente de la demande dépassé.

Statistiques Ping pour 100.64.0.10:
    Paquets : envoyés = 2, reçus = 0, perdus = 2 (perte 100%),
Ctrl+C
^C
C:\Users\Administrateur>ssh tp2@100.64.0.10
ssh: connect to host 100.64.0.10 port 22: Connection timed out
```



4- Testez les communications (ping et SSH) depuis le routeur R1 vers SExt ?

```
R1#ssh -l tp2 100.64.0.10
% Destination unreachable; gateway or host down

R1#ping 100.64.0.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.64.0.10, timeout is 2 seconds:
UUUUU
Success rate is 0 percent (0/5)
```

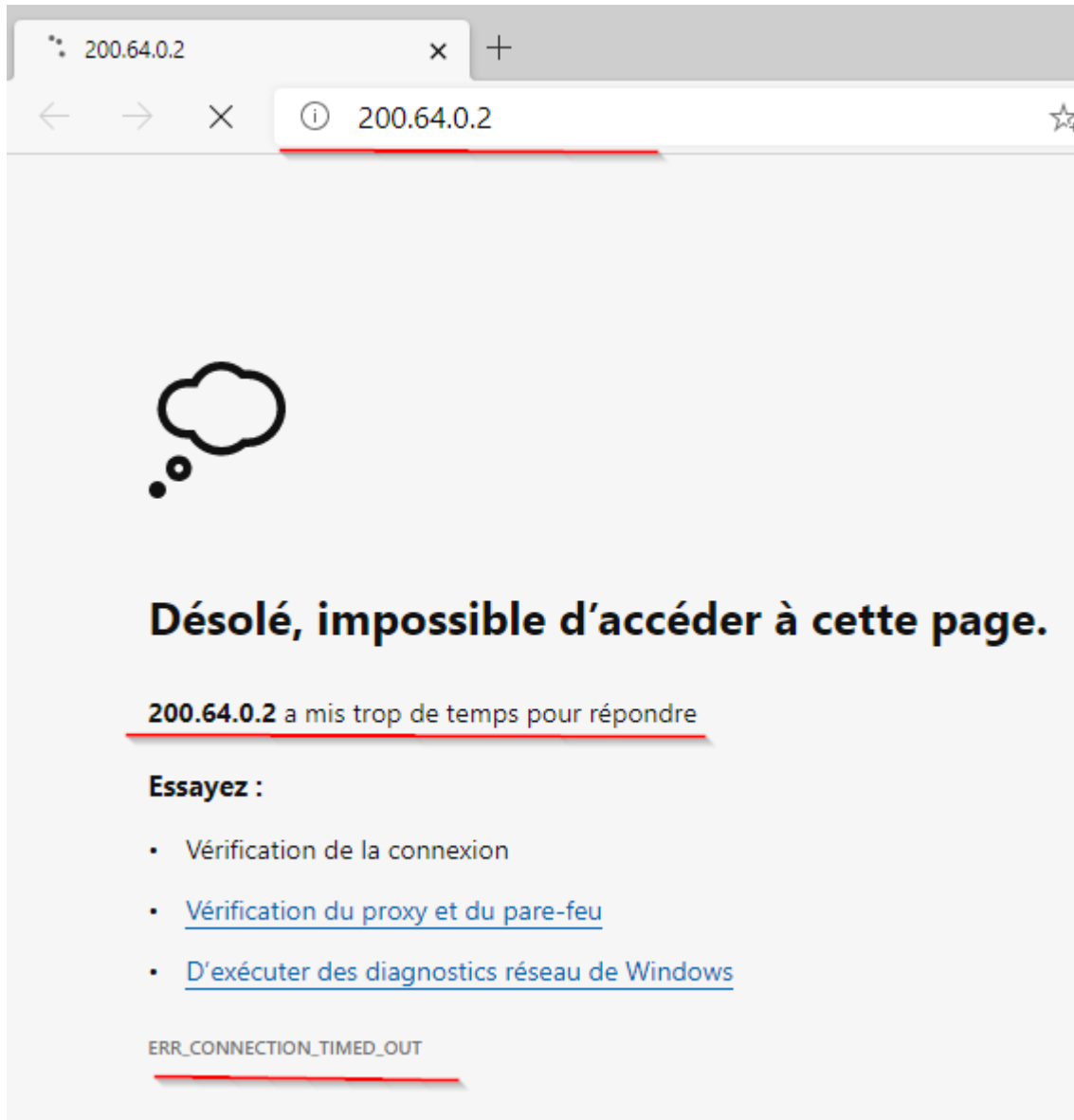
5- Testez les communications (ping, SSH et WEB) depuis la VM SExt vers la VM SDMZ ?

```
C:\Users\Administrateur>ping 200.64.0.2

Envoi d'une requête 'Ping' 200.64.0.2 avec 32 octets de données :
Réponse de 100.64.0.10 : Impossible de joindre l'hôte de destination.
Réponse de 100.64.0.10 : Impossible de joindre l'hôte de destination.
Réponse de 100.64.0.10 : Impossible de joindre l'hôte de destination.
Réponse de 100.64.0.10 : Impossible de joindre l'hôte de destination.

Statistiques Ping pour 200.64.0.2:
    Paquets : envoyés = 4, reçus = 4, perdus = 0 (perte 0%),

C:\Users\Administrateur>ssh tp2@200.64.0.2
ssh: connect to host 200.64.0.2 port 22: Connection timed out
```



6- Testez les communications (ping et SSH) depuis la VM SExt vers le routeur R1 ?


```
C:\Users\Administrateur>ping 10.2.64.1
```

```
Envoi d'une requête 'Ping' 10.2.64.1 avec 32 octets de données :  
Réponse de 100.64.0.10 : Impossible de joindre l'hôte de destination.  
Délai d'attente de la demande dépassé.  
Délai d'attente de la demande dépassé.  
Délai d'attente de la demande dépassé.
```

```
Statistiques Ping pour 10.2.64.1:
```

```
    Paquets : envoyés = 4, reçus = 1, perdus = 3 (perte 75%),
```

```
C:\Users\Administrateur>ssh tp2@10.2.64.1
```

```
ssh: connect to host 10.2.64.1 port 22: Connection timed out
```

Routeur Cisco avec ACL (5 points)

7- Configurez le routeur R1 avec du routage statique ?

```
R1(config)#interface GigabitEthernet3
R1(config-if)# ip address 10.1.64.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)#exit
R1(config)#
R1(config)#interface GigabitEthernet4
R1(config-if)# ip address 10.2.64.1 255.255.255.252
R1(config-if)# no shutdown
R1(config-if)#exit
R1(config)#
R1(config)#ip route 0.0.0.0 0.0.0.0 10.2.64.2

R1(config)#username tp2 privilege 15 secret Cmm02026!
R1(config)#enable secret Cmm02026!
R1(config)#
R1(config)#crypto key generate rsa modulus 2048
The name for the keys will be: R1.tp2.local

% The key modulus size is 2048 bits
% Generating 2048 bit RSA keys, keys will be non-exportable
[OK] (elapsed time was 0 seconds)

R1(config)#ip ssh version 2
R1(config)#
R1(config)#line vty 0 4
R1(config-line)# login local
R1(config-line)# transport input ssh
R1(config-line)#exit
R1(config)#
R1(config)#end
R1#write memory
```

8- Testez les communications (ping et WEB) vers Internet depuis le routeur R1 ?

```

R1#ping 100.64.0.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.64.0.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
R1#telnet 100.64.0.10 80
Trying 100.64.0.10, 80 ... Open
FHTTP/1.1 400 Bad Request
Content-Type: text/html; charset=us-ascii
Server: Microsoft-HTTPAPI/2.0
Date: Mon, 18 May 2026 19:43:28 GMT
Connection: close
Content-Length: 326

<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<HTML><HEAD><TITLE>Bad Request</TITLE>
<META HTTP-EQUIV="Content-Type" Content="text/html; charset=us-ascii"></HEAD>
<BODY><h2>Bad Request - Invalid Verb</h2>
<hr><p>HTTP Error 400. The request verb is invalid.</p>
</BODY></HTML>

[Connection to 100.64.0.10 closed by foreign host]
R1#_

```

9- Testez les communications (ping, SSH et WEB) depuis le routeur R1 vers SExt ?

```

R1#ping 100.64.0.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.64.0.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
R1#telnet 100.64.0.10 80
Trying 100.64.0.10, 80 ... Open
FHTTP/1.1 400 Bad Request
Content-Type: text/html; charset=us-ascii
Server: Microsoft-HTTPAPI/2.0
Date: Mon, 18 May 2026 19:43:28 GMT
Connection: close
Content-Length: 326

<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<HTML><HEAD><TITLE>Bad Request</TITLE>
<META HTTP-EQUIV="Content-Type" Content="text/html; charset=us-ascii"></HEAD>
<BODY><h2>Bad Request - Invalid Verb</h2>
<hr><p>HTTP Error 400. The request verb is invalid.</p>
</BODY></HTML>

[Connection to 100.64.0.10 closed by foreign host]
R1#_
R1#ssh -l tp2 100.64.0.10
Password:

Microsoft Windows [version 10.0.20348.1850]
(c) Microsoft Corporation. Tous droits réservés.

tp2@WIN-10VIE4A73VL C:\Users\tp2>_

```

10- Testez la communication (ping) depuis la machine Kali vers R1 ?

```
$ ping -c 4 10.1.64.1
```

```
PING 10.1.64.1 (10.1.64.1) 56(84) bytes of data.  
64 bytes from 10.1.64.1: icmp_seq=1 ttl=255 time=1.29 ms  
64 bytes from 10.1.64.1: icmp_seq=2 ttl=255 time=0.830 ms  
64 bytes from 10.1.64.1: icmp_seq=3 ttl=255 time=0.602 ms  
64 bytes from 10.1.64.1: icmp_seq=4 ttl=255 time=0.981 ms  
  
--- 10.1.64.1 ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3042ms  
rtt min/avg/max/mdev = 0.602/0.924/1.285/0.248 ms
```

Puis testez les communications (ping, SSH et WEB) de Kali vers le réseaux externe (SExt) ?

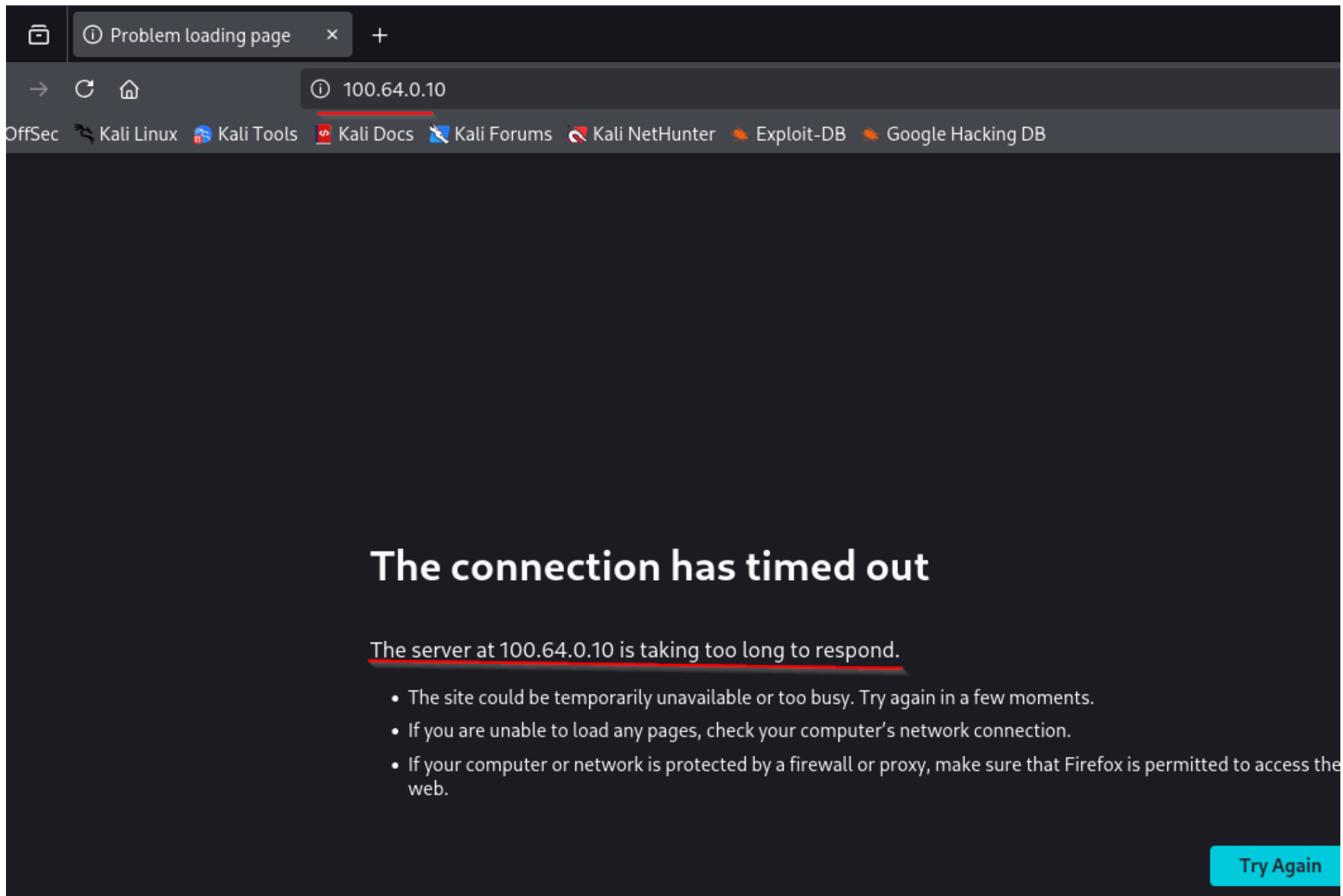
```
$ ping -c 4 100.64.0.10
```

```
PING 100.64.0.10 (100.64.0.10) 56(84) bytes of data.  
  
--- 100.64.0.10 ping statistics ---  
4 packets transmitted, 0 received, 100% packet loss, time 3067ms
```

```
(administrateur@kali)-[~]
```

```
$ ssh tp2@100.64.0.10
```

```
ssh: connect to host 100.64.0.10 port 22: Connection timed out
```



11- Configurez le protocole PAT sur le routeur R1, autoriser tout le LAN à utiliser le NAT ?

```

R1(config)#access-list 1 permit 10.1.64.0 0.0.0.255
R1(config)#
R1(config)#interface GigabitEthernet1
R1(config-if)# ip nat inside
R1(config-if)#exit
R1(config)#
R1(config)#interface GigabitEthernet2
R1(config-if)# ip nat outside
R1(config-if)#exit
R1(config)#
R1(config)#ip nat inside source list 1 interface GigabitEthernet2 overload

```

12- Retestez les communications (ping, SSH et WEB) de Kali vers le réseau externe (SExt) ?

```

$ ping -c 4 100.64.0.10

PING 100.64.0.10 (100.64.0.10) 56(84) bytes of data.
64 bytes from 100.64.0.10: icmp_seq=1 ttl=126 time=1.64 ms
64 bytes from 100.64.0.10: icmp_seq=2 ttl=126 time=2.59 ms
64 bytes from 100.64.0.10: icmp_seq=3 ttl=126 time=1.17 ms
64 bytes from 100.64.0.10: icmp_seq=4 ttl=126 time=1.20 ms

```

```

$ ssh tp2@100.64.0.10

tp2@100.64.0.10's password:
Microsoft Windows [version 10.0.20348.1850]
(c) Microsoft Corporation. Tous droits réservés.

tp2@WIN-10VIE4A73VL C:\Users\tp2>
(administrateur@kali) ~
$ curl http://100.64.0.10
SExt - Serveur externe 100.64.0.10

```

13- Configurez la politique de filtrage suivante sur le routeur R1 :

- Autorisez toutes les communications depuis le LAN Interne vers la DMZ ?
- Autorisez toutes les communications DNS, WEB et SSH depuis le LAN vers le réseau externe ?
- Autorisez les pings depuis la machine Kali vers la VM SExt ?
- Autorisez le protocole SSH vers R1 uniquement depuis la machine Kali ?

```

R1(config)#ip access-list extended LAN-FILTRE
R1(config-ext-nacl)# permit ip 10.1.64.0 0.0.0.255 200.64.0.0 0.0.0.7
R1(config-ext-nacl)#$ 10.1.64.0 0.0.0.255 100.64.0.0 0.0.255.255 eq domain
R1(config-ext-nacl)#$ 10.1.64.0 0.0.0.255 100.64.0.0 0.0.255.255 eq domain
R1(config-ext-nacl)#$ 10.1.64.0 0.0.0.255 100.64.0.0 0.0.255.255 eq www
R1(config-ext-nacl)#$ 10.1.64.0 0.0.0.255 100.64.0.0 0.0.255.255 eq 443
R1(config-ext-nacl)#$ 10.1.64.0 0.0.0.255 100.64.0.0 0.0.255.255 eq 22
R1(config-ext-nacl)# permit icmp host 10.1.64.11 host 100.64.0.10 echo
R1(config-ext-nacl)# deny ip 10.1.64.0 0.0.0.255 100.64.0.0 0.0.255.255
R1(config-ext-nacl)# permit ip any any
R1(config-ext-nacl)#exit
R1(config)#interface GigabitEthernet1
R1(config-if)# ip access-group LAN-FILTRE in
R1(config-if)#exit
R1(config)#
R1(config)#ip access-list standard ACCES-R1
R1(config-std-nacl)# permit 10.1.64.11
R1(config-std-nacl)# deny any
R1(config-std-nacl)#exit
R1(config)#line vty 0 4
R1(config-line)# access-class ACCES-R1 in
R1(config-line)# login local
R1(config-line)# transport input ssh
R1(config-line)#exit
R1(config)#

```

14- Testez la politique de filtrage :

- Testez ping et SSH depuis Kali vers le routeur R1 ?

```

$ ping -c 4 10.1.64.1
PING 10.1.64.1 (10.1.64.1) 56(84) bytes of data:
64 bytes from 10.1.64.1: icmp_seq=1 ttl=255 time=0.518 ms
64 bytes from 10.1.64.1: icmp_seq=2 ttl=255 time=0.938 ms
64 bytes from 10.1.64.1: icmp_seq=3 ttl=255 time=0.693 ms
64 bytes from 10.1.64.1: icmp_seq=4 ttl=255 time=0.937 ms

```



```

$ ssh tp2@10.1.64.1
The authenticity of host '10.1.64.1 (10.1.64.1)' can't be established.
RSA key fingerprint is: SHA256:zopKuAknNGVl3PE2+QlWmYxDRmpyi09l9yGFRofmi2s
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.1.64.1' (RSA) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
(tp2@10.1.64.1) Password:

```

R1#

b. Testez ping, SSH et WEB depuis Kali vers la VM SExt ?

```

$ ping -c 4 100.64.0.10

PING 100.64.0.10 (100.64.0.10) 56(84) bytes of data:
64 bytes from 100.64.0.10: icmp_seq=1 ttl=126 time=0.709 ms
64 bytes from 100.64.0.10: icmp_seq=2 ttl=126 time=2.18 ms
64 bytes from 100.64.0.10: icmp_seq=3 ttl=126 time=2.61 ms
64 bytes from 100.64.0.10: icmp_seq=4 ttl=126 time=2.08 ms

--- 100.64.0.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3032ms
rtt min/avg/max/mdev = 0.709/1.894/2.605/0.711 ms

```

```

Microsoft Windows [version 10.0.20348.1850]
(c) Microsoft Corporation. Tous droits réservés.

```

```

tp2@WIN-10VIE4A73VL C:\Users\tp2>

```

```

(administrateur@kali)-[~]
$ curl http://100.64.0.10
SExt - Serveur externe 100.64.0.10

```

c. Changez l'adresse IP de la machine Kali avec l'adresse 10.X.1.12 ?

The screenshot shows the 'Filaire' (Wireless) network configuration window in Kali Linux. The 'IPv4' tab is selected. Under 'Méthode IPv4', the 'Manuel' (Manual) option is selected. The 'Adresses' (Addresses) table shows the IP address '10.1.64.12' with a subnet mask of '255.255.255.0' and a gateway of '10.1.64.1'. The 'DNS' section shows the 'Automatique' (Automatic) toggle is turned on, and the DNS server address is '100.64.0.10'.

Adresse	Masque de réseau	Passerelle
10.1.64.12	255.255.255.0	10.1.64.1

DNS: 100.64.0.10

d. Testez ping et SSH depuis Kali vers le routeur R1 ?

```
$ ping -c 4 10.1.64.1

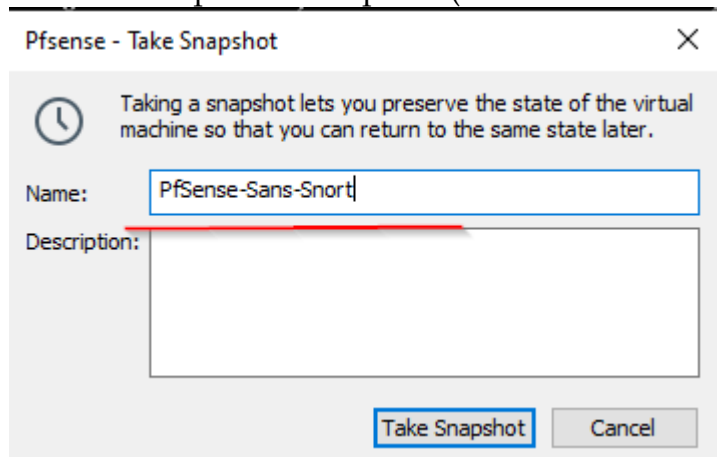
PING 10.1.64.1 (10.1.64.1) 56(84) bytes of data:
64 bytes from 10.1.64.1: icmp_seq=1 ttl=255 time=0.197 ms
64 bytes from 10.1.64.1: icmp_seq=2 ttl=255 time=0.520 ms
64 bytes from 10.1.64.1: icmp_seq=3 ttl=255 time=0.344 ms
64 bytes from 10.1.64.1: icmp_seq=4 ttl=255 time=0.954 ms

$ ssh tp2@10.1.64.1
ssh: connect to host 10.1.64.1 port 22: Connection refused
```

e. Testez ping, SSH et WEB depuis Kali vers la VM SExt ?

```
(administrateur@kali)-[~]  
$ ping -c 4 100.64.0.10  
  
PING 100.64.0.10 (100.64.0.10) 56(84) bytes  
From 10.1.64.1 icmp_seq=1 Packet filtered  
From 10.1.64.1 icmp_seq=2 Packet filtered  
From 10.1.64.1 icmp_seq=3 Packet filtered  
From 10.1.64.1 icmp_seq=4 Packet filtered  
  
--- 100.64.0.10 ping statistics ---  
4 packets transmitted, 0 received, +4 error  
  
Microsoft Windows [version 10.0.20348.1850]  
(c) Microsoft Corporation. Tous droits réservés.  
  
tp2@WIN-10VIE4A73VL C:\Users\tp2>  
  
$ curl http://100.64.0.10  
SExt - Serveur externe 100.64.0.10
```

15- Faites un premier snapshot (PfSense-Sans-Snort) de la machine PfSense ?



Faites un deuxième snapshot (PfSense-Avec-Snort) de la machine PfSense ?

Pfsense - Take Snapshot ✕

 Taking a snapshot lets you preserve the state of the virtual machine so that you can return to the same state later.

Name:

Description:

Snort avec PfSense (2 points)

16- Sur le Snapshot PfSense-Avec-Snort, mettez l'interface WAN en mode 'bridged' ?

Virtual Machine Settings

Hardware Options

Device	Summary
Memory	1 GB
Processors	1
Hard Disk (IDE)	8 GB
CD/DVD (IDE)	Using file C:\Users\Administra...
Network Adapter	Custom (VMnet0)
Network Adapter 2	LAN Segment
Network Adapter 3	LAN Segment
Sound Card	Auto detect
Display	Auto detect

Device status

- ☒ Connected
☒ Connect at power on

Network connection

- ☐ Bridged: Connected directly to the physical network
☐ Replicate physical network connection state
- ☐ NAT: Used to share the host's IP address
- ☐ Host-only: A private network shared with the host
- ☒ Custom: Specific virtual network
- VMnet0 (Bridged) ▼
- ☐ LAN segment:
- ▼

LAN Segments...

Advanced...

Add...

Remove

OK

Cancel

Help

17- Vérifiez que vous avez l'accès à Internet ?

General Configuration

Enable ☒ Enable interface

Description

WAN

Enter a description (name) for the interface here.

IPv4 Configuration Type

DHCP

IPv6 Configuration Type

None

MAC Address

xx:xx:xx:xx:xx:xx

This field can be used to modify ("spoof") the MAC address of this interface.

Enter a MAC address in the following format: xx:xx:xx:xx:xx:xx or leave blank.

MTU

If this field is blank, the adapter's default MTU will be used. This is typically 1500 bytes but can vary in some circumstances.

MSS

If a value is entered in this field, then MSS clamping for TCP connections to the value entered above minus 40 for IPv4 (TCP/IPv4 header size) and minus 60 for IPv6 (TCP/IPv6 header size) will be in effect.

Speed and Duplex

Default (no preference, typically autoselect)

Explicitly set speed and duplex mode for this interface.

WARNING: MUST be set to autoselect (automatically negotiate speed) unless the port this interface connects to has its speed and duplex forced.

DHCP Client Configuration

Options

☐ Advanced Configuration

Use advanced DHCP configuration options.

☐ Configuration Override

Override the configuration from this file.

Hostname

The value in this field is sent as the DHCP client identifier and hostname when requesting a DHCP lease. Some ISPs may require this (for client identification).

Alias IPv4 address

/ 32

The value in this field is used as a fixed alias IPv4 address by the DHCP client.

Reject leases from

To have the DHCP client reject offers from specific DHCP servers, enter their IP addresses here (separate multiple entries with a comma). This is useful for rejecting leases from cable modems that offer private IP addresses when they lose upstream sync.

Reserved Networks

Block private networks and loopback addresses

☐

Blocks traffic from IP addresses that are reserved for private networks per RFC 1918 (10/8, 172.16/12, 192.168/16) and unique local addresses per RFC 4193 (fc00::/7) as well as loopback addresses (127/8). This option should generally be turned on, unless this network interface resides in such a private address space, too.

Block bogon networks

☐

```
WAN (wan)      -> em0      -> v4/DHCP4: 10.64.103.181/23
LAN (lan)      -> em1      -> v4: 10.2.64.2/30
DMZ (opt1)     -> em2      -> v4: 200.64.0.1/29

0) Logout (SSH only)      9) pfTop
1) Assign Interfaces      10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults 13) Update from console
5) Reboot system         14) Enable Secure Shell (sshd)
6) Halt system           15) Restore recent configuration
7) Ping host             16) Restart PHP-FPM
8) Shell

Enter an option: 2
```

18- Installez et configurez le package de détection d'intrusion **Snort** sur PfSense ?

pfSense-pkg-snort installation successfully completed.

[Installed Packages](#)[Available Packages](#)[Package Installer](#)

Package Installation

Please note that, by default, snort will truncate packets larger than the default snaplen of 15158 bytes. Additionally, LRO may cause issues with Stream5 target-based reassembly. It is recommended to disable LRO, if your card supports it.

This can be done by appending '-lro' to your ifconfig_ line in rc.conf.

=====

Message from pfSense-pkg-snort-4.1.6_17:

--

Please visit Services - Snort - Interfaces tab first to add an interface, then select your desired rules packages at the Services - Snort - Global tab. Afterwards visit the Updates tab to download your configured rulesets.

>>> Cleaning up cache... done.

Success

WAN Interface (wan, em0)	
Status	up 
DHCP	up  Release WAN ☐ Relinquish Lease
MAC Address	00:0c:29:b7:d1:4b
IPv4 Address	10.64.103.181
Subnet mask IPv4	255.255.254.0
Gateway IPv4	10.64.102.1
IPv6 Link Local	fe80::20c:29ff:feb7:d14b%em0
DNS servers	10.64.0.2 10.64.0.242 10.64.0.243 10.64.0.244
MTU	1500
Media	1000baseT <full-duplex>
In/out packets	4443/5397 (535 KiB/275 KiB)
In/out packets (pass)	4443/5397 (535 KiB/275 KiB)
In/out packets (block)	1719/0 (125 KiB/0 B)
In/out errors	0/0
Collisions	0
Interrupts	15592 (1/s)

19- Faites un exemple de test d'intrusion avec Snort depuis la machine Kali ?

```
(administrateur@kali)-[~]
└─$ sudo nmap -sS -p 1-102 10.64.103.181
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-19 00:53 EDT
Nmap scan report for 10.64.103.181
Host is up (0.00078s latency).
Not shown: 100 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http

Nmap done: 1 IP address (1 host up) scanned in 14.90 seconds
```

WARNING: The 'admin' account password is set to the default value. [Change the password in the User Manager.](#)

Services / Snort / Alerts



Snort Interfaces Global Settings Updates **Alerts** Blocked Pass Lists Suppress IP Lists SID Mgmt Log Mgmt Sync

Alert Log View Settings

Interface to Inspect

WAN (em0) ▾

☒ Auto-refresh view

250

Save

Choose interface..

Alert lines to display.

Alert Log Actions



Alert Log View Filter



7 Entries in Active Log

Date	Action	Pri	Proto	Class	Source IP	SPort	Destination IP	DPort	GID:SID	Description
2026-05-18 18:54:36		2		Attempted Information Leak	fe80::be24:11ff:fec2:1ff		ff02::fb		122:24	(portscan) UDP Filtered Distributed Portscan
2026-05-18 18:54:36		2		Attempted Information Leak	10.64.102.16		224.0.0.251		122:24	(portscan) UDP Filtered Distributed Portscan
2026-05-18 18:52:55		2		Attempted Information Leak	fe80::be24:11ff:fec2:5bc4		ff02::fb		122:24	(portscan) UDP Filtered Distributed Portscan
2026-05-18 18:52:55		2		Attempted Information Leak	10.64.102.20		224.0.0.251		122:24	(portscan) UDP Filtered Distributed Portscan
2026-05-18 18:52:45		2		Attempted Information Leak	10.64.102.146		151.101.126.172		122:7	(portscan) TCP Filtered PortswEEP
2026-05-18 18:51:19		2		Attempted Information Leak	fe80::be24:11ff:fe3c:fd4		ff02::fb		122:24	(portscan) UDP Filtered Distributed Portscan
2026-05-18 18:51:19		2		Attempted Information Leak	10.64.102.15		224.0.0.251		122:24	(portscan) UDP Filtered Distributed Portscan